

UESTCHN3007 Dynamics & Control
Lecture 1

Overview of Dynamic Systems

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Systems

Sets of connected objects or things

They can be living things

They can be mechanical things

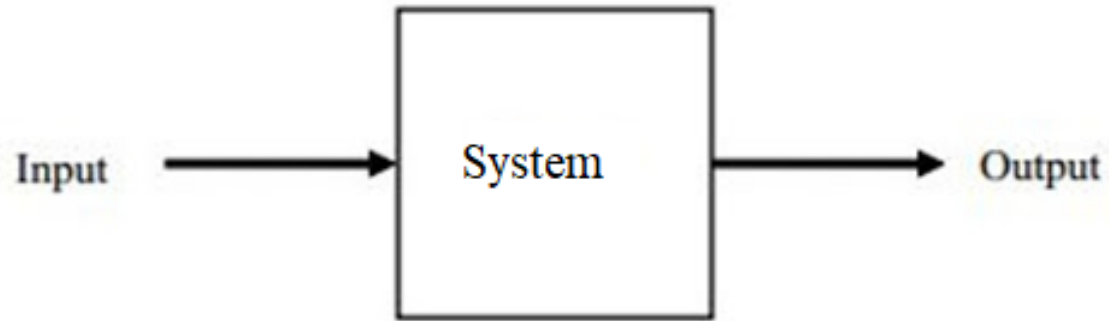
Behaviors of these systems are shaped by many factors

All these systems exhibit some common behavior patterns

What is a Dynamic System?

- A dynamic system can be set in motion by applying an **input** (or excitation) or by disturbing it from its **equilibrium**.
- The motion of the system is described by its **output** (or response).
- The **output** of the system is dependent on the **input** and the **behaviour** of the system itself.
- If we know what the behaviour of the system is, we can specify an **input** so that we get a desired **output**.
- But how do we describe the behaviour of the system?

The output of the system is dependent on the input and the behaviour of the system itself.

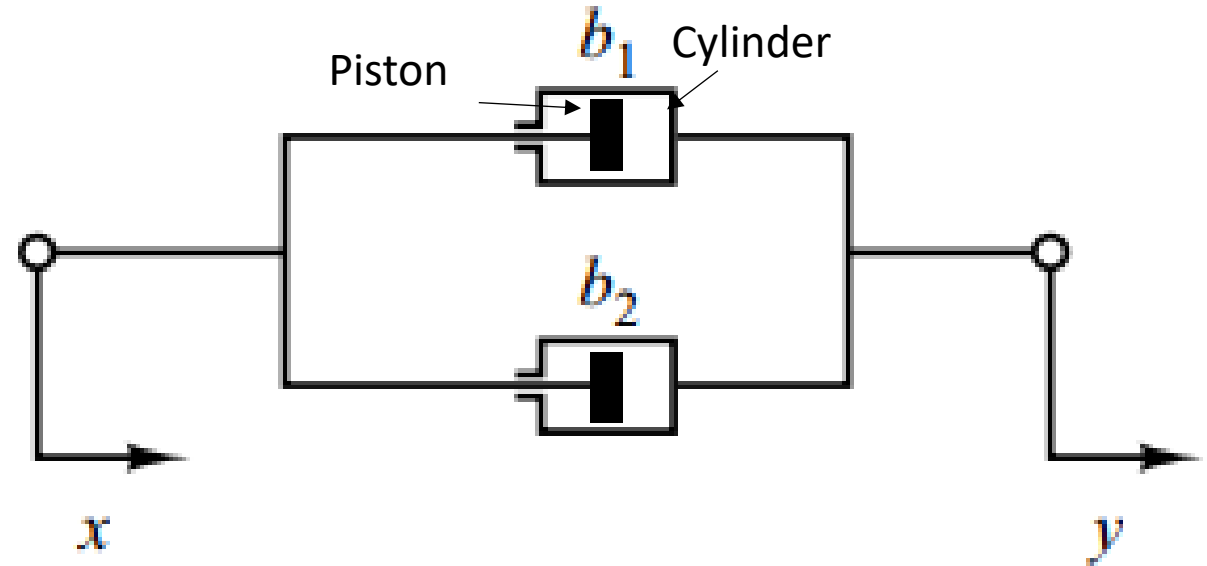


I/P and O/P relationship

- Model dynamic systems in mathematical terms (Why?)
- A set of equations that represents the dynamics of the system
- System may have many mathematical models
- The dynamics of systems may be described in terms of differential equations
- Use physical laws governing a particular system
- Testing a prototype of the device, or measuring its response to inputs
- Deriving reasonable mathematical models is the most important part

Mechanical System Modelling

System consisting of two dampers connected in parallel

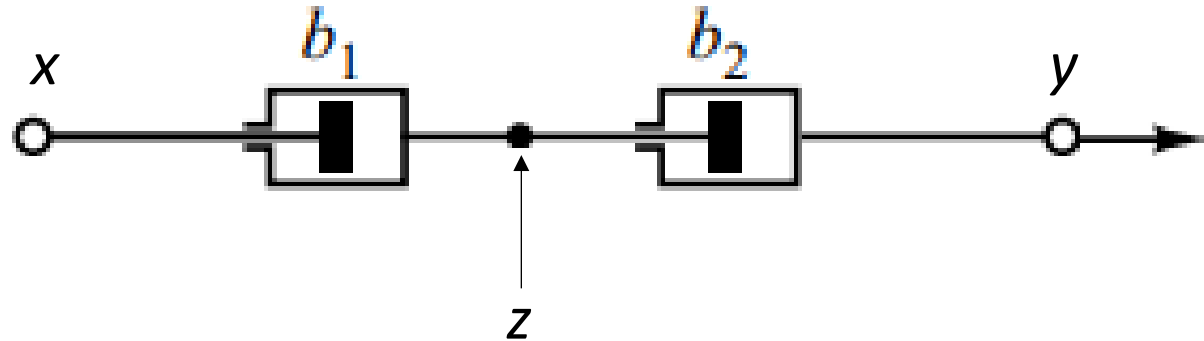


b_1 & b_2 are viscous-friction coefficients
 x and y are positions. The force F due to the dampers is

$$F = b_1(\dot{y} - \dot{x}) + b_2(\dot{y} - \dot{x})$$

Exercise: Mechanical System Modelling

System consisting of two dampers connected in series



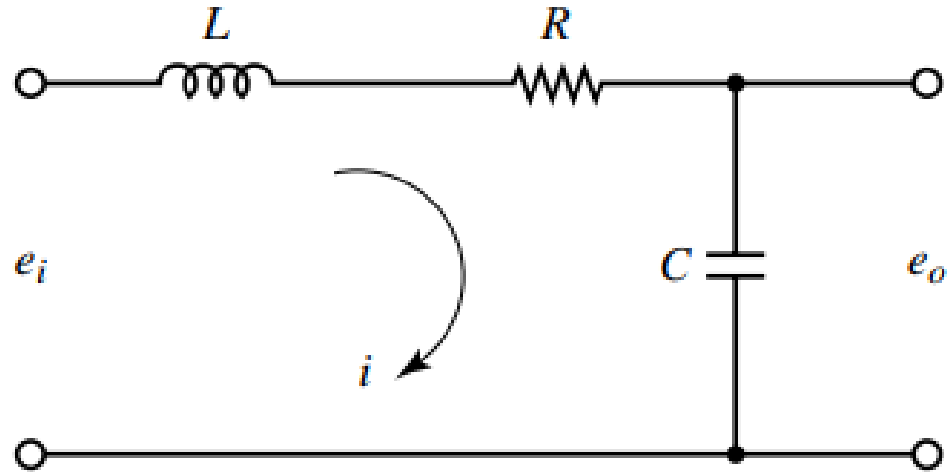
b_1 & b_2 are viscous-friction coefficients
 x and y are positions. The force F due to the dampers is

$$F = b_1(\dot{z} - \dot{x}) = b_2(\dot{y} - \dot{z})$$



Electrical System Modelling

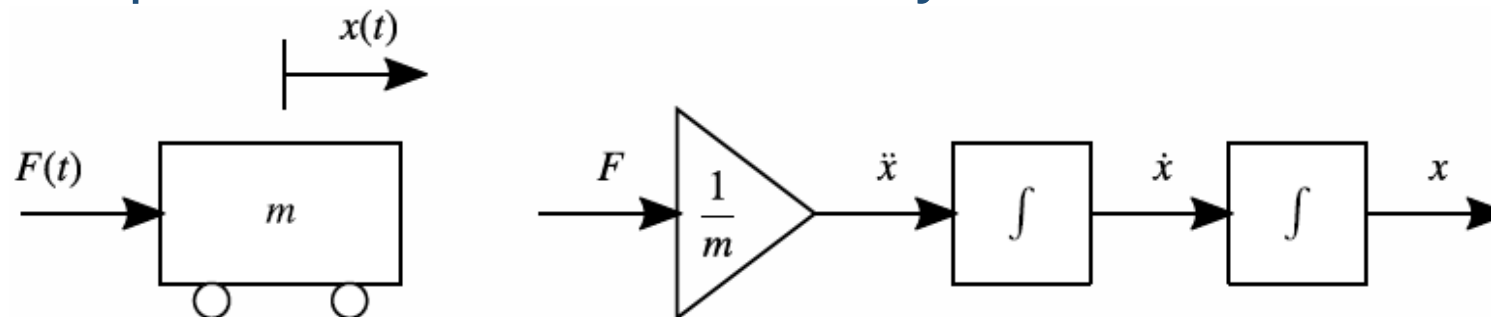
Consider the electrical circuit as shown below



$$L \frac{di}{dt} + Ri + \frac{1}{C} \int i dt = e_i$$

A Mechanical System

- A simple trolley is subject to an **input** force $F(t)$.
- From Newton's 2nd Law, we know
- $F(t) = m\ddot{x}(t)$
- where m is the (constant) mass of the trolley and $\ddot{x}(t)$ is its acceleration, or the second derivative of position. The **state/output** is then the position $x(t)$.
- As the **second** derivative of the output is proportional to the input, this is a **second-order** system.



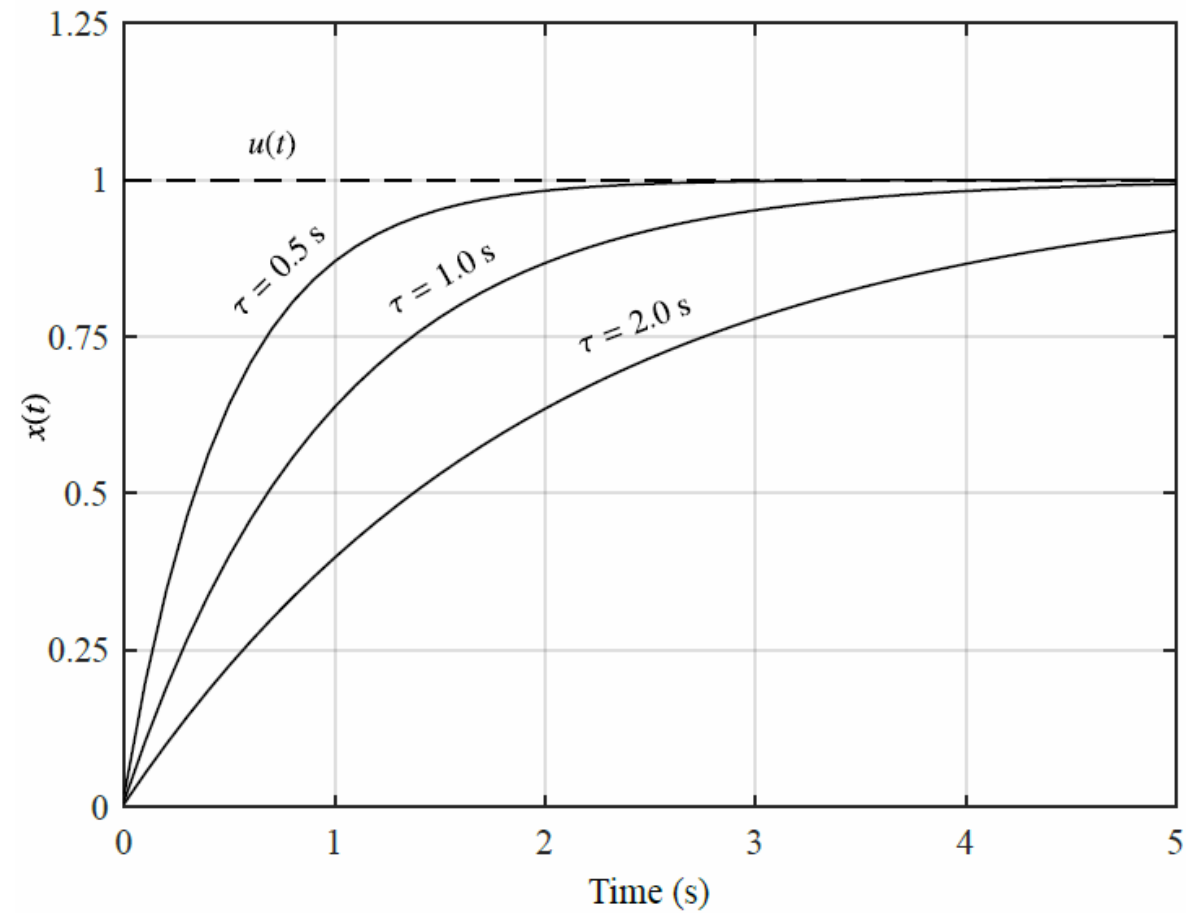
Describing a Dynamic System

- Equations describing the systems known as system's equation of motion.
- Describes the O/P of the system
- Describes its derivatives as a function of the I/P
- Can have MIMO

First-Order System

$$\dot{x}(t) + \frac{1}{\tau}x(t) = Ku(t)$$

- where $x(t)$ is the state/output, $u(t)$ is the input, τ is the time constant and K is the input gain.
- Varying τ changes the “speed” of the system, while varying K changes the output magnitude with respect to the input.

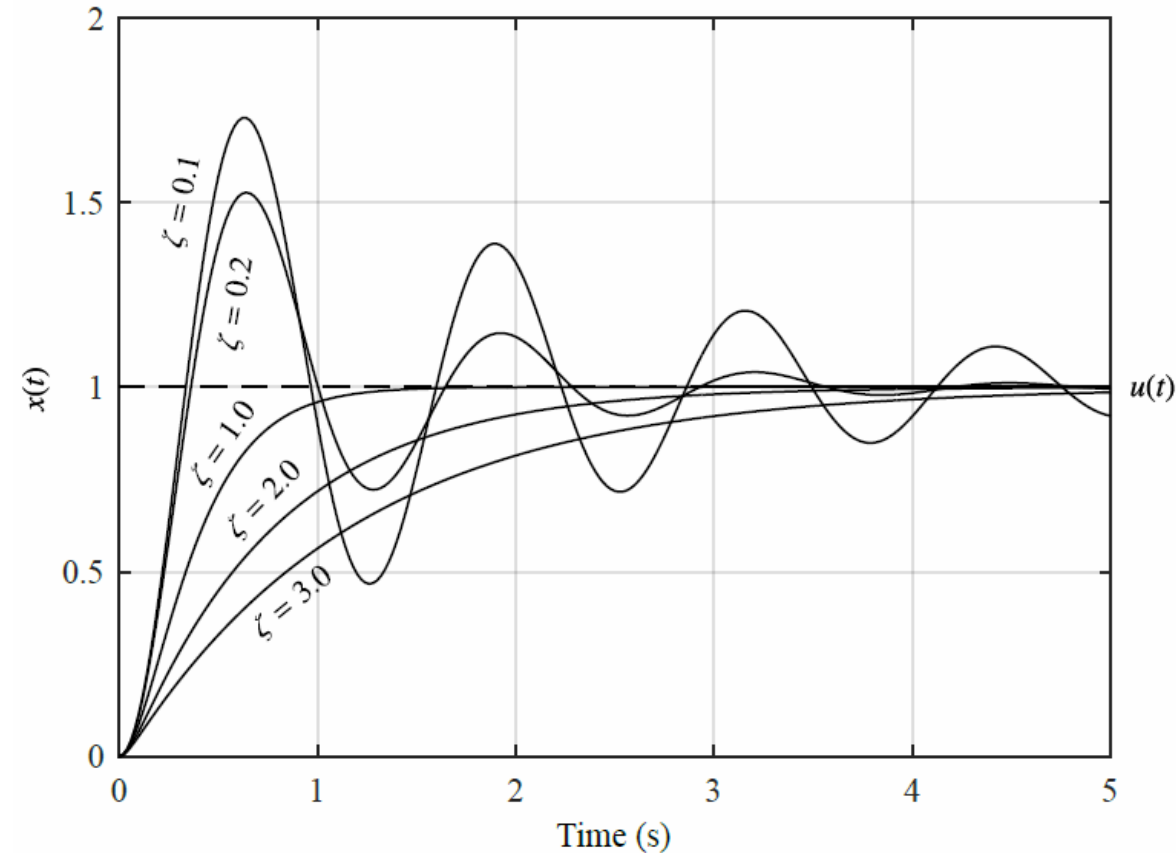


Second-Order System

$$\ddot{x}(t) + 2\zeta\omega_n\dot{x}(t) + \omega_n^2x(t) = Ku(t)$$

where $x(t)$ is the **state/output**,
 $u(t)$ is the **input**, ζ is the **damping ratio**, ω_n is the **natural frequency** and K is the input gain.

Varying ζ changes how much the system oscillates, while varying ω_n changes the “speed” of the system.



Higher-Order Systems

An n^{th} - order system has the general equation of motion

$$x^{(n)}(t) + a_1 x^{(n-1)}(t) + a_2 x^{(n-2)}(t) + \cdots + a_n x(t) = Ku(t)$$

The Laplace Transform

$$\mathcal{L}[f(t)] = F(s) = \int_{0-}^{\infty} f(t)e^{-st} dt$$

- Differential and integral equations are difficult to model as a block diagram
- The input, output, and system are represented separately with their own equations
- Their interrelationship will be represented as algebraic equations

$f(t)$	$F(s)$
$\delta(t)$	1
$u(t)$	$\frac{1}{s}$
$tu(t)$	$\frac{1}{s^2}$
$t^n u(t)$	$\frac{n!}{s^{n+1}}$
$e^{-at}u(t)$	$\frac{1}{s+a}$
$\sin \omega t u(t)$	$\frac{\omega}{s^2 + \omega^2}$
$\cos \omega t u(t)$	$\frac{s}{s^2 + \omega^2}$

Transfer Functions

- Describe the relationship between the input (I/P) and the output (O/P)

$$\dot{x}(t) + \frac{1}{\tau}x(t) = Ku(t)$$

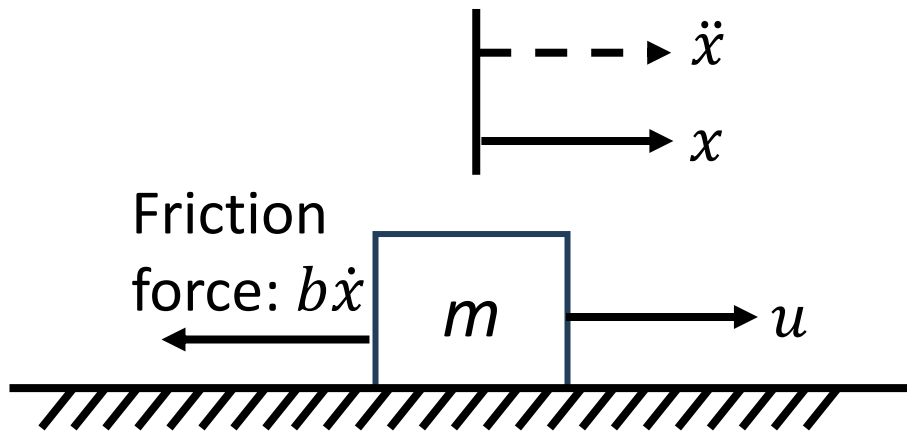
- Using the Laplace domain, we get

$$sx(s) + \frac{1}{\tau}x(s) = Ku(s)$$

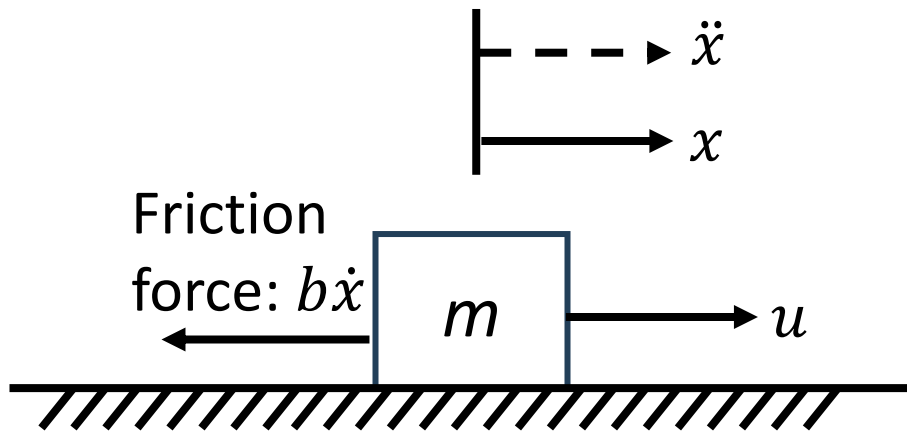
$$\frac{x(s)}{u(s)} = \frac{K}{s + \frac{1}{\tau}}$$

Exercise: Transfer Function for a mechanical system

- Applying Newton's law, $F = m \cdot a$
- F is the vector sum of all forces
- a vector acceleration in m/s^2
- m mass of the body in kg
- A force $u(t)$ is applied to a mass m and an opposing force of friction applies



Exercise: Transfer Function for a mechanical system



- $F = m \cdot a$
- $u - b\dot{x} = m\ddot{x}$
- $\ddot{x} + \frac{b}{m}\dot{x} = \frac{u}{m}$
- Converting using the Laplace transforms
- $s^2X(s) + \frac{b}{m}sX(s) = \frac{1}{m}U(s)$
- $ms^2X(s) + bsX(s) = U(s)$

$$\frac{X(s)}{U(s)} = \frac{1}{ms^2 + bs}$$

Obtain transfer functions for the below systems

Second-order system: $\ddot{x}(t) + 2\zeta\omega_n\dot{x}(t) + \omega_n^2x(t) = Ku(t)$

n^{th} -order system: $x^{(n)}(t) + a_1x^{(n-1)}(t) + a_2x^{(n-2)}(t) + \cdots + a_nx(t) = Ku(t)$

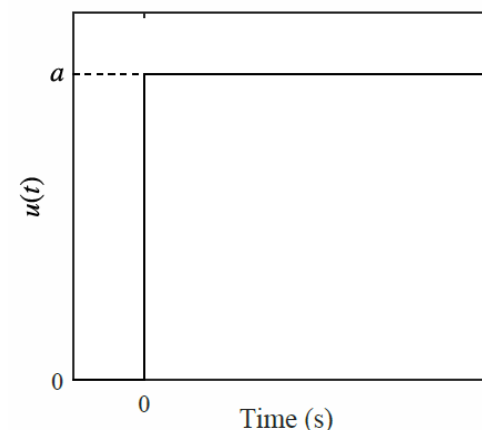
Types of Input

- The objective of a controller is to determine the input $u(t)$ for a specific response.
- However, $u(t)$ can also be a constant or any time-dependent function. By applying a certain input and observing the output response, we can learn much about the system.

The most popular input is the **step input**, defined by

$$u(t) = 0 \text{ for } t < 0$$

$$u(t) = a \text{ for } t \geq 0$$



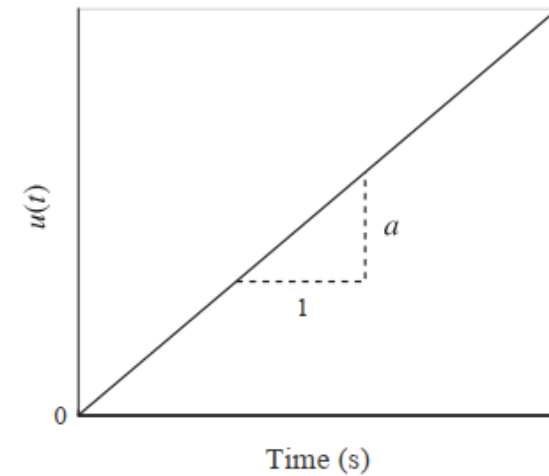
$$\frac{a}{s}$$

Types of Input

The **ramp input**, defined by

$$u(t) = 0 \text{ for } t < 0$$

$$u(t) = at \text{ for } t \geq 0$$



$$\frac{a}{s^2}$$

Response Properties

- By applying a step input to a system which is at rest, we cause the system to be in forced vibration.
- The system will be in motion before settling to a constant **steady-state value**. The phase where the system is in motion is called the **transient phase**. In theory, the steady-state phase occurs at $t = \infty$, although in practice it occurs much sooner.
- The steady-state value may be found using the final-value theorem. For a second-order system, the final-value theorem yields
- $$x_{ss} = x(\infty) = \frac{Ka}{\omega_n^2}$$

Response Properties etc.

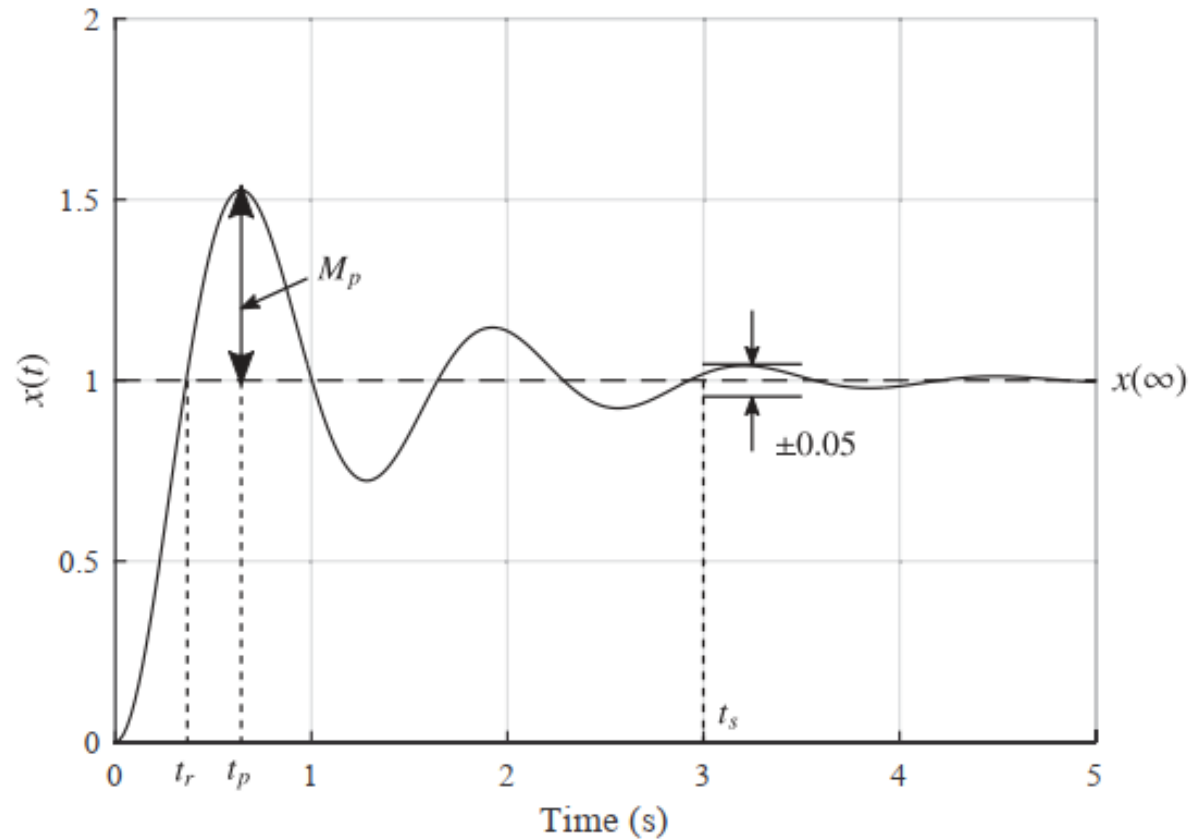
The response of a system can be described by several properties

Maximum peak value of the response: M_p

Rise time: t_r

Peak time: t_p

Settling time: t_s



Controlling a System

- These properties only hold if a system is **stable**. An **unstable** system will never reach a steady-state value after receiving an input, or will never reach an equilibrium after being disturbed.
- Some systems, such as fighter aircraft, are deliberately unstable. In these cases, we design a **control system** which stabilises the aircraft.
- We can design our control system so that it gives our complete a system a desired property or set of properties.

Summary of Section 1

- A dynamic system can be set in motion by applying an **input** or moving the system away from its **equilibrium**
- A dynamic system can take many forms, with some properties being:
 - The **order** of the system, $1, 2, 3, \dots n$
 - The **stability** of the system
 - The shape of the **transient** response
 - The properties of the **steady-state** response
- We can use a **controller** to change the properties of the system by regulating its **input**
- This means we can:
 - Make an unstable system stable
 - Increase the order of a system
 - Change the damping and speed of a system response

Summary

- Basics of Dynamic Systems
- Mathematical Models
- Transfer Function
- Response Characteristics

References:

-*Modern Control Engineering, 5th Edition, K. Ogata*
-*UESTC3001 2019/20 Notes, J. Le Kernec etc.*